NASA Battery Aging Dataset Battery Health Prediction Using Classical ML & Neural Networks

Introduction

This report represents the implemented ML project using the NASA Battery Aging dataset. Our implementation is divided into 2 main phases: Phase 1 focused on the statistical implementation covering data exploration and hypothesis testing, whereas Phase 2 focused on implementing ML models covering feature extraction, dimension reduction, applying supervised & unsupervised learning models using both Regression & Classification models.

The dataset shows how 34 lithium-ion batteries degrade over time under various ambient temperatures. Each of the 2,769 records represents a single charge-discharge cycle, with measurements of voltage, current, temperature, discharge duration, and derived health indicators like State of Health (SOH) and Remaining Useful Life (RUL). It contains detailed time-series measurements from 34 lithium-ion batteries cycled to end-of-life at different ambient temperatures (4°C, 22°C, 24°C, 43°C, and 44°C). This temperature variation is particularly interesting because it allows us to study how thermal conditions influence degradation, something that turns out to matter quite a lot.

The achieved results in this project as follows:

Project Phases	Approach	Key Result
Phase 1: Statistical Analysis	EDA, ANOVA, t-test, Confidence Intervals	Temperature & health class both significantly affect capacity ($p < 0.05$)
Phase 2a: Feature Engineering + PCA	6 engineered features + StandardScaler + PCA	30 clean features reduced to 16 PCs (95% variance retained)
Phase 2b: Clustering	K-Means + HDBSCAN	K=2 clusters align perfectly with Healthy/Degraded classes
Phase 2c: Regression (SOH)	Ridge, Polynomial, SVR	Ridge Regression achieved $R^2 = 0.88$ on test set
Phase 2d: Classification (Health Class)	SVM + ANN (MLP)	SVM: 95.4% accuracy + ANN: 93.6% accuracy

Problem Statement

The project addresses two fundamental prediction problems:

- **Regression:** Predict the State of Health (SOH) of a battery cycle as a continuous value between 0 and 1, where 1 represents a brand-new battery.
- **Classification:** Determine whether a battery cycle belongs to the 'Healthy' or 'Degraded' class.

Dataset Description

The dataset consists of 2,769 rows and 59 columns, spanning cycle-level summaries of electrical and thermal measurements. After preprocessing, the usable feature space contains 52 clean columns. Key features in the dataset:

- Capacity (Ah): the total charge delivered during discharge, the primary indicator of health
- SOH: State of Health, computed as current capacity divided by the initial rated capacity
- RUL_cycles: Remaining Useful Life, how many more cycles the battery can complete before failing
- Voltage_measured_mean/std/min/max: statistical summaries of the discharge voltage profile
- Current_measured_mean/std: summaries of the discharge current
- Temperature_measured_mean/std/max: thermal behaviour during the cycle
- cycle_number: the age of the battery in cycles
- ambient_temperature: the temperature of the test chamber
- health_class: binary label (Healthy / Degraded) based on an SOH threshold.

Phase 1: Statistical Analysis and Data Exploration

1. Dataset Shape:
   - 2,769 rows × 59 columns
   - 34 unique batteries
   - Ambient temperatures: 4°C, 22°C, 24°C, 43°C, 44°C
2. The columns Re (electrolyte resistance) and Rct (charge transfer resistance) were entirely missing, with all 2,769 values set to NaN. Instead of imputing 100% of the values as missing (which would be manufacturing information out of thin air), these columns were dropped. Also, other columns were removed: source_file, uid, start_time, initial_capacity, and SOH_relative. These were either identifiers, timestamps, or derived duplicates of other columns. After this cleanup, the dataset was fully clean with zero missing values across all 52 remaining features.

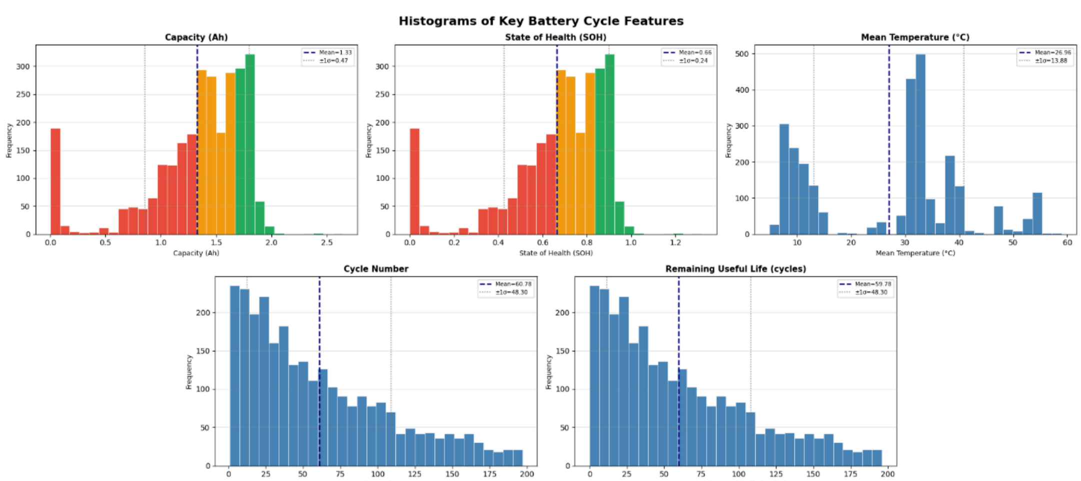
Numerical Summary Statistics

Feature	Mean	Std Dev
Capacity (Ah)	1.3265	0.4725
SOH	0.79 (approx)	0.23 (approx)
Voltage_measured_mean (V)	3.4313	0.1985
Current_measured_mean (A)	-1.7288	0.7815
Temperature_measured_mean (°C)	26.9613	13.8797
cycle_number	~70	~55

The wide standard deviation on Capacity (0.47 Ah on a mean of 1.33 Ah, or about 35%) already signals that this dataset captures a large range of health states, from new batteries near 2.0 Ah to heavily degraded ones approaching 0 Ah. The Current_measured_mean is negative because discharge current flows out of the battery, a sign convention maintained consistently throughout. The Temperature_measured_mean shows huge variability (std = 13.88°C) because the dataset aggregates batteries tested in chambers ranging from 4°C to 44°C. Within a single temperature group, thermal variability is much smaller.

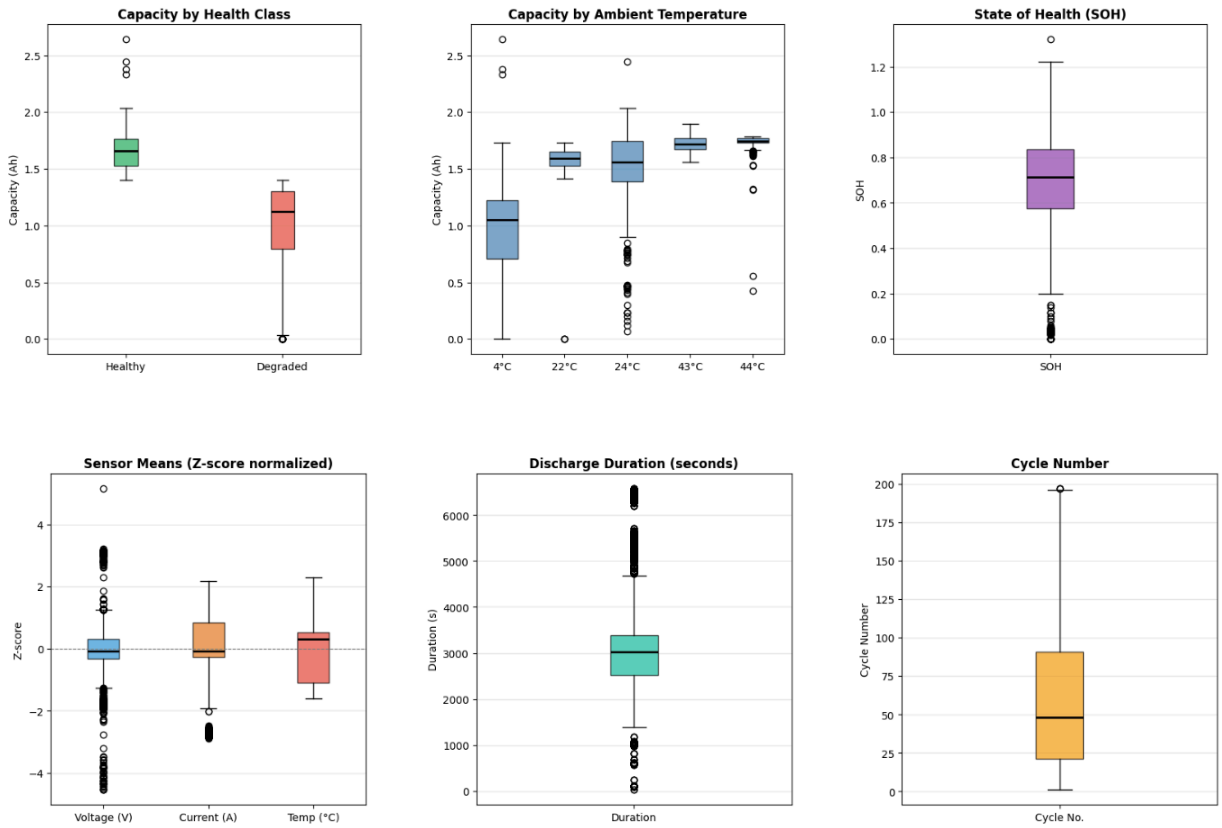
Distribution Analysis and Outlier Detection

Histograms of Capacity, SOH, Temperature, Cycle Number, and RUL were plotted to visually separate early-life, mid-life, and end-of-life states. The distributions reveal a bimodal trend in Capacity and SOH, with clusters corresponding to the Healthy and Degraded populations, which would later be confirmed by clustering.



Box plots were used to detect outliers across the dataset using the IQR method. The results were:

- **Capacity:** 214 outliers (7.7%) identified, primarily cycles at temperature extremes (4°C cold-degraded and 43°C heat-stressed batteries)
- **SOH:** 214 outliers (7.7%), matching Capacity since SOH is directly derived from it

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Outlier Summary (IQR method)
 Capacity                         → 214 outliers (7.7%)
 SOH                              → 214 outliers (7.7%)
 Voltage_measured_mean            → 352 outliers (12.7%)
 Current_measured_mean            → 206 outliers (7.4%)
 Temperature_measured_mean        →   0 outliers (0.0%)
 duration_seconds                 → 651 outliers (23.5%)
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These outliers were not removed. Battery degradation is inherently non-linear, and the extreme-value cycles carry fair engineering information about failure modes. Removing them would bias the model toward healthy operation and undermine its predictive value at the decision boundary.

Statistical Inference

ANOVA: Effect of Ambient Temperature on Capacity

Used ANOVA to test whether the ambient temperature of the test chamber has a statistically significant effect on battery capacity.

Temperature Group	Mean Capacity (Ah)	Sample Size
4°C	0.9040	1,006
22°C	1.5512	123
24°C	1.5407	1,375
43°C	1.7212	160
44°C	1.7053	105

ANOVA Result;

- F-statistic = 606.43
- p-value ≈ 0.000000
- Decision: REJECT H_0 — Temperature significantly affects Capacity ($p < 0.05$)

With an F-statistic of 606, this is not a close call. Temperature has a strong effect on capacity. Also, cold-temperature batteries (4°C) have substantially lower average capacity (0.90 Ah) compared to warmer groups (1.55 Ah for room temperature, 1.72 Ah for elevated temperature).

Welch's t-Test: Effect of Health Class on Capacity

A two-sample Welch's t-test compared the Capacity distributions of the Healthy and Degraded classes. Welch's variant is appropriate here because the two groups have substantially different variances (as we would expect from healthy vs. failing batteries).

Health Class	Mean Capacity (Ah)	Std Dev (Ah)
Healthy	1.6504	0.1466
Degraded	0.9569	0.4441

t-Test Result

- t-statistic = 53.64
- p-value ≈ 0.000000

- Decision: REJECT H_0 — Health class significantly affects Capacity

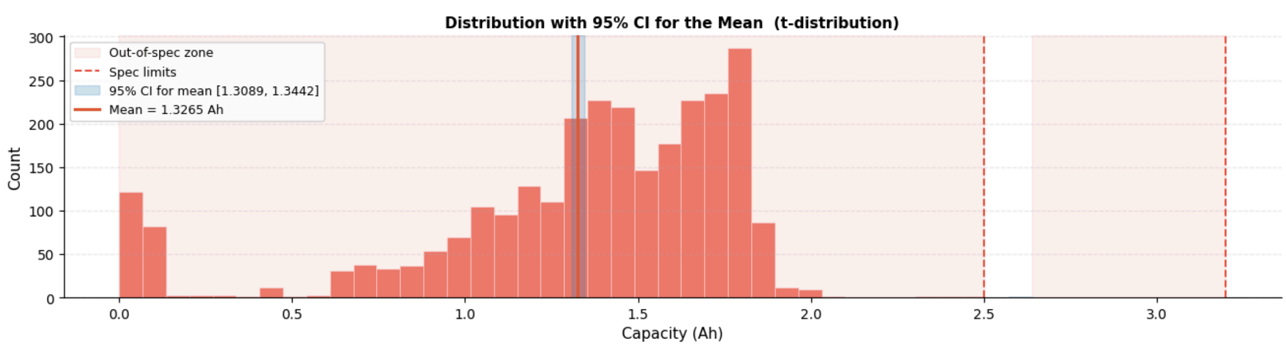
The capacity gap between Healthy (1.65 Ah) and Degraded (0.96 Ah) batteries is nearly 0.70 Ah, and the t-statistic of 53.64 is huge. This confirms what engineering instinct would predict: the health label is a very strong predictor of capacity. The wider standard deviation for Degraded cells (0.44 vs 0.15) also reflects the heterogeneity of failure: batteries degrade at different rates and in different ways depending on usage history and temperature.

Confidence Interval Analysis

A 95% confidence interval was computed for the mean, variance, and standard deviation of Capacity using a t-distribution for the mean and a chi-squared distribution for the variance:

Statistic	Estimate	95% CI
Mean Capacity (Ah)	1.32654	[1.30894, 1.34415]
Variance	0.22327	[0.21196, 0.23552]
Std Dev (Ah)	0.47252	[0.46039, 0.48530]

The confidence interval for the mean is narrow (width = 0.035 Ah), reflecting the large sample size (n = 2,769). However, the engineering risk assessment flagged this as HIGH risk because the mean capacity of 1.33 Ah falls well outside the typical specification band of 2.5-3.2 Ah for modern cells.



Phase 2a: Feature Engineering and Dimensionality Reduction

Feature Engineering

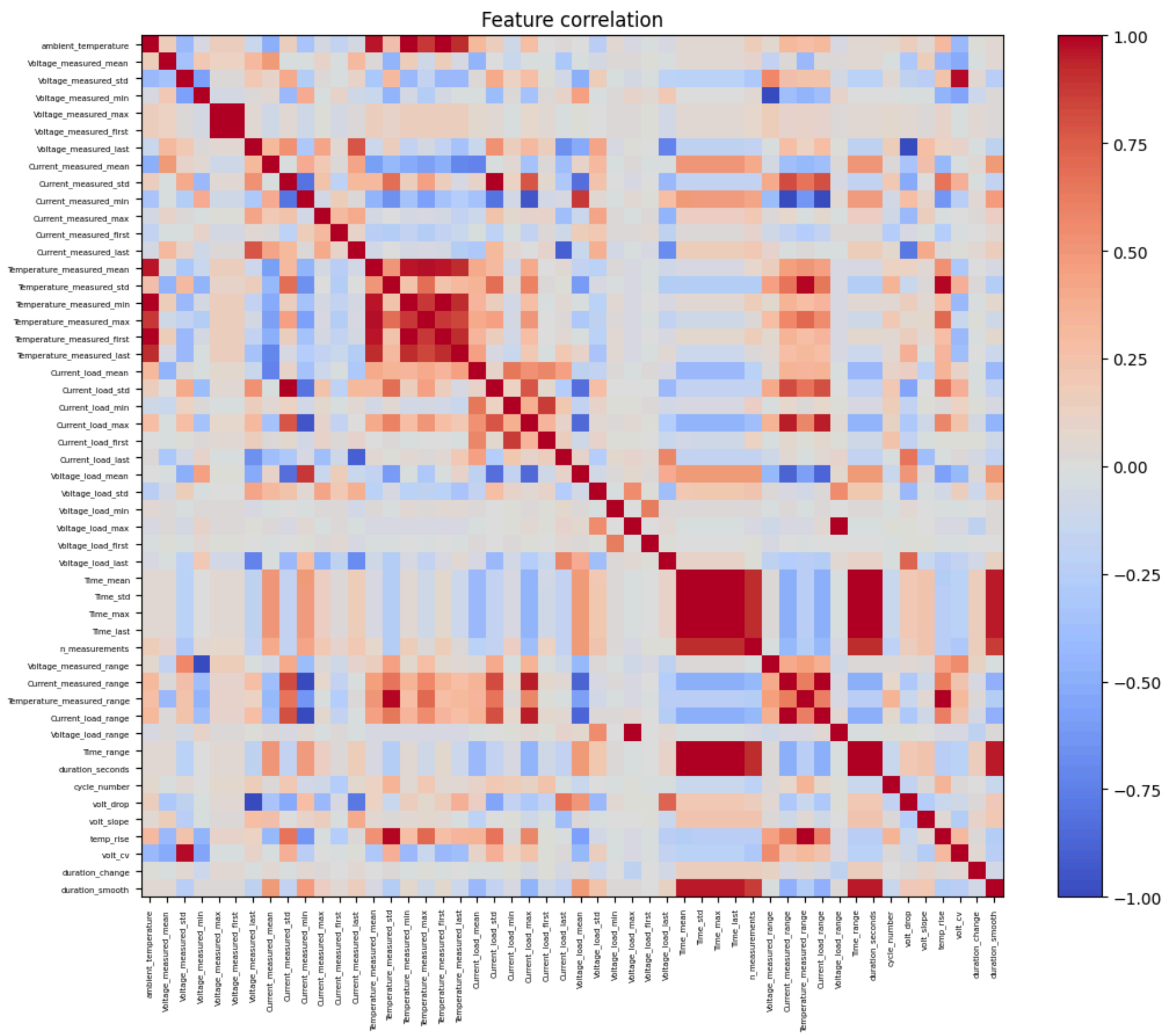
Raw sensor readings from battery cycles are informative but not always the most efficient representation for learning algorithms. A voltage reading at a single moment in time tells you less than the rate at which voltage is falling over the entire cycle. Six new features were engineered to capture dynamics that the raw statistics miss:

- **volt_drop:** Voltage_measured_first minus Voltage_measured_last. Measures how far the terminal voltage fell during the discharge. A large drop indicates a cell struggling to maintain voltage, which is a sign of degradation.
- **volt_slope:** (Voltage_measured_last - Voltage_measured_first) / duration_seconds. The rate of voltage decline per second, normalised for cycle duration. This is essentially a proxy for internal resistance growth.
- **temp_rise:** Temperature_measured_max minus Temperature_measured_first. How much the battery heated up during discharge. Higher temperature rise can indicate increased internal resistance and Joule heating in degraded cells.
- **volt_cv:** Voltage_measured_std / Voltage_measured_mean. The coefficient of variation of the voltage signal captures how erratically the voltage behaved relative to its mean level.
- **duration_change:** Cycle-over-cycle difference in discharge duration per battery. A battery that is discharging for progressively shorter periods is losing capacity.
- **duration_smooth:** Rolling 5-cycle average of discharge duration per battery. Smooths out measurement noise to reveal the underlying trend in discharge time.

Feature Selection and Cleaning

Starting with 59 raw columns:

- **Step 1: Drop entirely missing columns:** Re and Rct (both 100% NaN) were removed, leaving 57 columns plus the 6 engineered features.
- **Step 2: Separate targets:** Capacity, SOH, SOH_relative, initial_capacity, RUL_cycles, and health_class were set aside as targets, not inputs.
- **Step 3: Median imputation:** Remaining NaN values were filled with column medians, a robust strategy that avoids influence from outliers.
- **Step 4: Drop constant columns:** Time_min and Time_first had zero variance (same value in every row) and were removed as they carry no predictive signal.
- **Step 5: Drop near-duplicate columns:** A pairwise correlation matrix identified 20 columns with greater than 95% correlation with at least one other column. One from each redundant pair was dropped. This step is important because highly correlated features do not add information but do add noise to distance-based algorithms.



Feature Count Summary

- Raw features: 59
- After target removal: 54
- After constant/empty drops: 52
- After correlation filtering: 30 final features

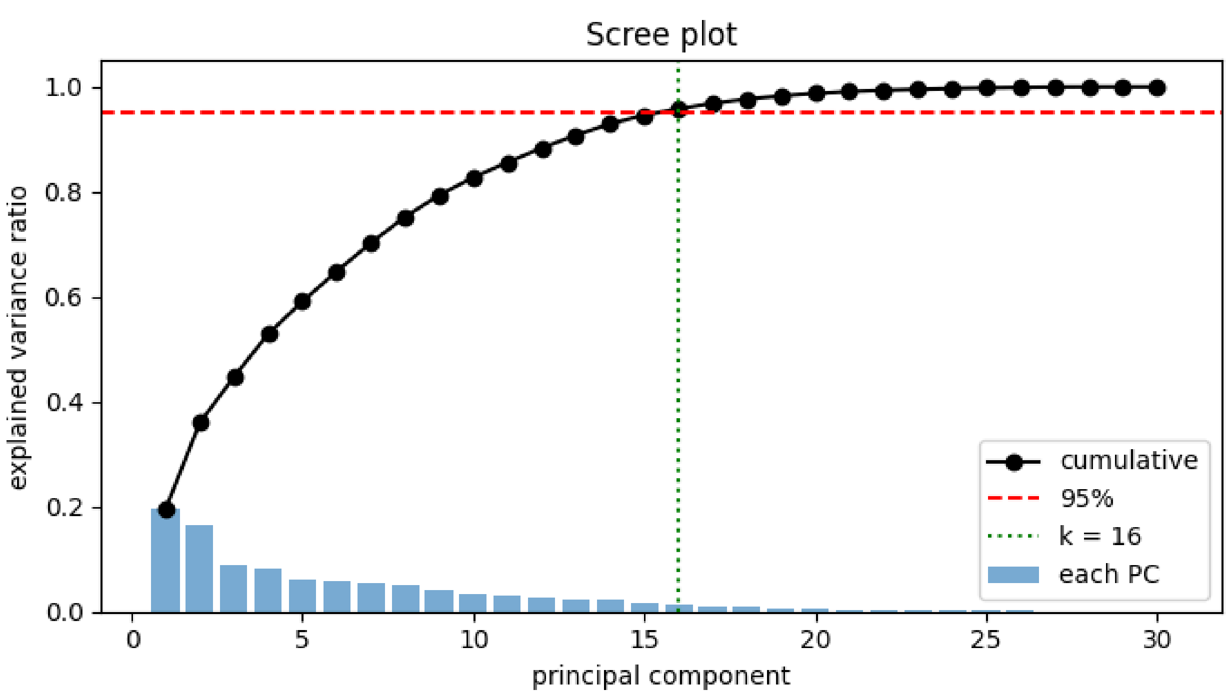
Principal Component Analysis (PCA)

PCA works by finding the directions of maximum variance in the feature space and projecting the data onto those directions. Because PCA is sensitive to feature scale (a feature measured in seconds would dominate one measured in millivolts), the 30 features were standardised using StandardScaler before applying PCA. This ensures each feature contributes equally based on its variance, not its units.

Full PCA was fitted to the 30-dimensional standardised feature space. The explained variance ratio by principal component was:

PC	Individual Variance	Cumulative Variance
PC1	19.6%	19.6%
PC2	16.4%	36.0%
PC3	8.8%	44.8%
PC4	8.2%	53.0%
PC5	6.2%	59.2%
PC6-PC10	~5% each	~80%
PC11-PC16	2-3% each	95.0%

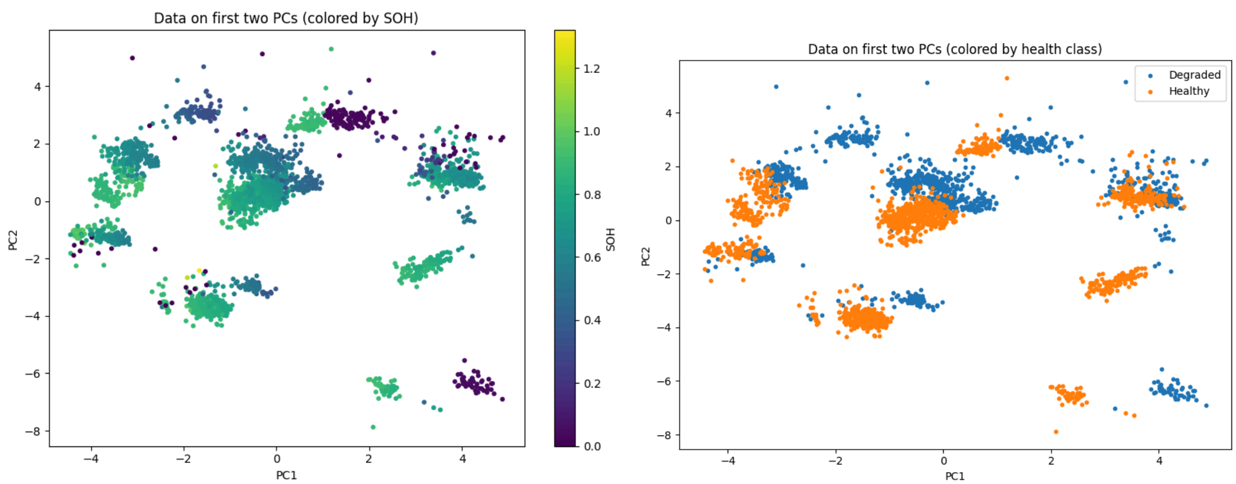
The scree plot showed that reaching 95% explained variance requires 16 principal components, meaning that despite starting with 30 features, the data effectively lives in a 16-dimensional subspace. This is a significant but not dramatic reduction, reflecting the fact that battery cycle features, while partially correlated, capture genuinely different aspects of degradation.



An examination of PCA1 & PCA2 shows which original features most strongly define the first two components:

- **PC1 (19.6% variance):** Dominated by Current_measured_min (-0.373) and Voltage_load_mean (-0.364) on the negative end, and Current_load_max (+0.379) and Current_measured_std (+0.332) on the positive end. PC1 appears to capture the discharge energy and current draw pattern, essentially an axis of battery load intensity.
- **PC2 (16.4% variance):** Driven by Current_measured_last (+0.385) and Current_load_last (-0.363), with Voltage_measured_last (+0.357) and Voltage_load_last (-0.281). PC2 captures the end-of-discharge state, how much current the battery is still delivering, and at what voltage as it approaches cutoff.

When the data is plotted in the PC1-PC2 plane and coloured by SOH, the colour gradient flows smoothly from healthy (high SOH) to degraded (low SOH), confirming that PCA has successfully captured the ageing axis. The health_class overlay shows that the two classes are largely separable in this reduced space, which is excellent news for the subsequent classification models.



Phase 2b: Unsupervised Learning - Clustering

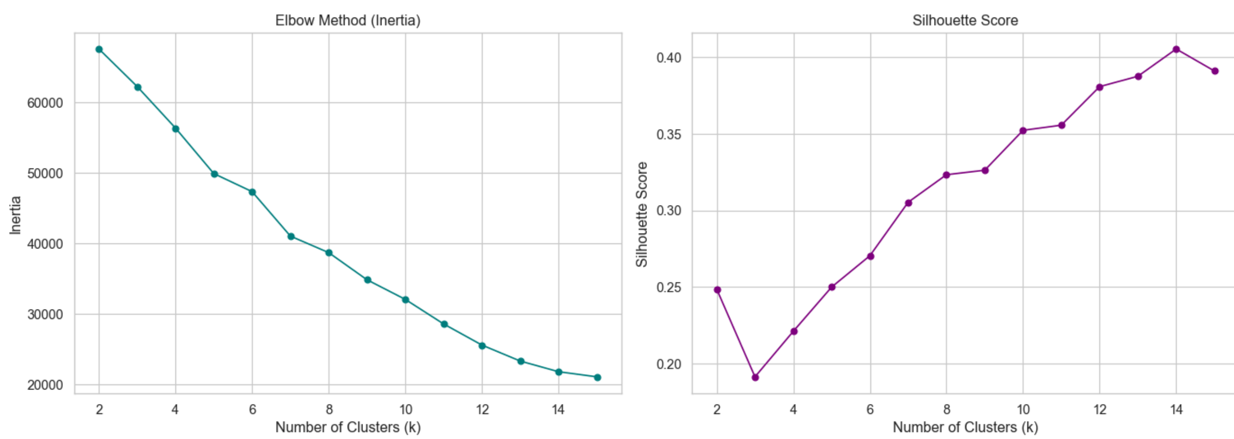
Before training supervised models, it is worth asking: Does the data organise itself into meaningful groups without any label guidance? This is the question unsupervised clustering answers. If the battery cycles cluster into groups that happen to align with the health classes we care about, that is independent validation that the structure is real and not an artefact of how the labels were constructed. The 16-dimensional PCA feature space was used as input for three clustering approaches: K-Means, and HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise).

Determining the Optimal Number of Clusters

K-Means requires you to specify the number of clusters k in advance. To find the right value, K was swept from 2 to 15, tracking two metrics at each step:

- **Inertia (WCSS, Elbow Method):** Within-cluster sum of squared distances from each point to its centroid. Drops sharply at first, then levels off. The elbow, where the rate of decrease slows down, marks the natural cluster count.
- **Silhouette Score:** Measures how similar each point is to its own cluster versus the nearest other cluster. Ranges from -1 to +1, with higher values meaning tighter, better-separated clusters.

Both metrics pointed clearly to k=2. The inertia curve showed its sharpest elbow at k=2, and the silhouette score peaked at k=2, confirming that the data has two dominant, natural groupings. This is exactly what we would expect given the binary Healthy/Degraded classification.



K-Means Implementation

A K-Means algorithm was implemented. The algorithm follows the standard Expectation-Maximization structure:

- **Initialization:** K centroids are randomly selected from the data points (random state fixed for reproducibility).
- **Assignment step:** Each data point is assigned to the nearest centroid using Euclidean distance.
- **Update step:** Each centroid is recalculated as the mean of all points assigned to it.
- **Convergence check:** If the sum of squared centroid shifts falls below tolerance (1e-4), training stops.

HDBSCAN Density-Based Clustering

HDBSCAN takes a different approach. Rather than assuming spherical clusters and requiring you to specify k upfront, it identifies regions of high density as clusters and labels low-density boundary points as noise. This makes it much more powerful for datasets with irregular cluster shapes and varying densities.

HDBSCAN Results

- Clusters found: 16
- Noise points: 211 (7.62%)
- The algorithm identified fine-grained density regions within what K-Means saw as two broad blobs

The 16 HDBSCAN clusters likely correspond to battery-specific degradation paths (different batteries degrade differently depending on their cycling history and temperature). The 7.62% noise label is also valuable: these are the boundary cycles that are ambiguous in their health state, exactly the cases where uncertainty-aware predictions would be most useful in a real system.

Cluster Evaluation

Metric	Sklearn K-Means (K=2)
Silhouette Score	0.2486
Davies-Bouldin Index	1.7044
Calinski-Harabasz Index	493.71

The silhouette score of 0.25 indicates moderate cluster separation, which makes sense: the Healthy and Degraded populations overlap somewhat in the boundary region, especially for batteries transitioning between states. HDBSCAN achieved a higher silhouette score (0.43) because its 16 clusters are smaller, tighter groupings, but its Calinski-Harabasz score (280) is lower, reflecting that the clusters have less separation relative to their within-cluster spread. K-Means correctly identified the underlying structure at K=2, corroborating the binary health classification. HDBSCAN revealed that the data is not simply two blobs, but contains richer substructure, possibly corresponding to individual battery degradation trajectories or temperature-specific behaviour. The 7.62% noise points represent the most uncertain, transitional cycles. Perhaps more importantly, the clustering provides independent validation that the Healthy/Degraded split is not arbitrary but reflects genuine structure in the electrical measurements.

Phase 2c: Supervised Learning - Regression

- SOH is the most clinically useful measure of battery health. It is a continuous value between 0 and 1, where 1.0 means the battery still has its full rated capacity and 0.0 means it is effectively dead. Being able to predict SOH from a single cycle's measurements enables real-time health monitoring without needing to discharge the battery to its cutoff limit. Before modelling, 206 rows with SOH ≤ 0.1 were dropped (these correspond to measurement failures or end-of-life cycles where the battery essentially stopped functioning). This left 2,563 usable records.

- Data Split
  - Training: 2,237 rows (from 27 batteries)
  - Test: 326 rows (from 7 batteries)
  - CV: 5-fold GroupKFold

Model 1: Ridge Regression

- Ridge regression is regularised linear regression that adds an L2 penalty to the coefficient magnitudes. This prevents any single feature from dominating the prediction and keeps the model stable when features are correlated (as they are here). The regularisation strength alpha was selected by cross-validation over the grid [0.1, 1, 10, 100].

- Ridge Regression Result
  - Best alpha: 100
  - Train RMSE: 0.0388
  - Val RMSE: 0.0644
  - Test RMSE: 0.0524
  - Test R^2: 0.8811
  - Test MAE: 0.0332
  - Verdict: Balanced — no real overfitting

- An R^2 of 0.88 means the model explains 88% of the variance in SOH on batteries it has never seen. The RMSE of 0.052 means predictions are typically off by about 5 SOH percentage points, which is excellent for a linear model with no domain-specific priors. The train-test gap (RMSE rising from 0.039 to 0.052) is small and expected, confirming the model is not overfitting.

Model 2: Polynomial Regression

- Polynomial regression extends linear regression by creating interaction and power terms from the features. To keep the feature expansion manageable (higher degree polynomials can generate thousands of terms from 30 features), the pipeline first reduced the data to 5 PCA components before applying polynomial expansion. The degree and alpha were jointly optimised by cross-validation.

- Polynomial Regression Result

- Best params: degree=1, alpha=10
- Train RMSE: 0.0868
- Test RMSE: 0.0757
- Test R^2: 0.7516
- Verdict: Balanced — no overfitting

The cross-validation notably selected degree=1, meaning the polynomial model collapsed to regularised linear regression on 5 PCs. This reveals something interesting: within the 5-PC subspace, the relationship is already linear. Adding polynomial terms from a reduced representation did not improve performance, suggesting the non-linearities in the data require either the full 16-PC space (as Ridge uses) or a non-linear kernel.

Model 3: Support Vector Regression (SVR)

- SVR with an RBF (Radial Basis Function) kernel maps the features into a high-dimensional space where complex non-linear relationships become linear. It then fits a regression hyperplane in that space, with an epsilon-insensitive loss function that ignores errors smaller than epsilon. Parameters C, gamma, and epsilon were tuned by cross-validation.

- SVR Result

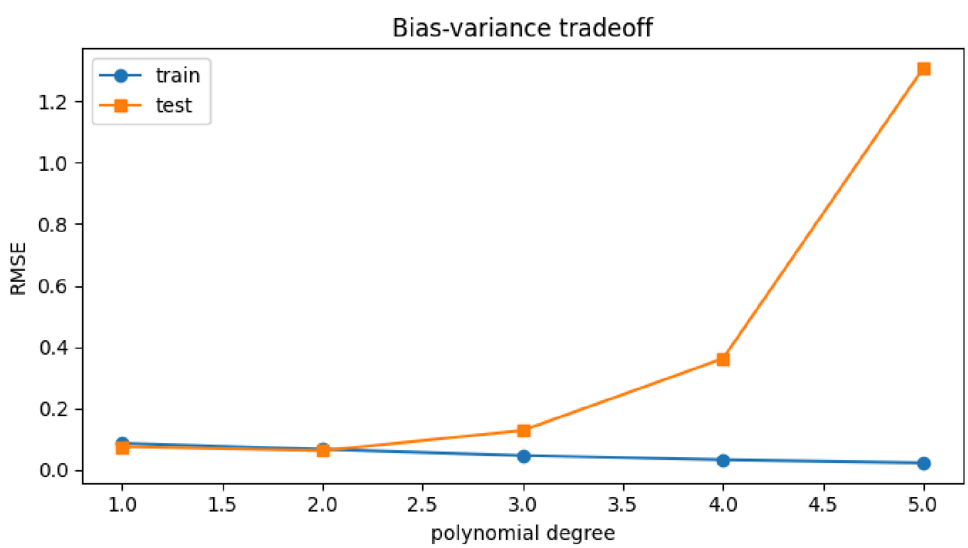
- Best params: C=1, epsilon=0.05, gamma='scale'
- Train RMSE: 0.0263
- Test RMSE: 0.0738
- Test R^2: 0.7642
- Verdict: Some overfitting

- SVR shows a meaningful train-test gap (RMSE goes from 0.026 to 0.074), flagged as 'some overfitting.' The very low training error suggests the RBF kernel fits the training cycles closely, but this sharpness did not fully transfer to unseen batteries. This is a known challenge with SVR in battery prognostics: each battery has its own idiosyncratic degradation path, making cross-battery generalisation harder for kernel methods that fit local geometry tightly.

Regression Results Comparison

Metric	Ridge	Polynomial
Train RMSE	0.0388	0.0868
Val RMSE	0.0644	0.0961
Test RMSE	0.0524	0.0757
Test MAE	0.0332	0.0534
Test MAPE (%)	5.63%	9.52%
Train R^2	0.9418	0.7081
Test R^2	0.8811	0.7516
Overfitting	None	None

Ridge Regression is the highest among the three regression models. It achieves the best test R^2 (0.88), the lowest test RMSE (0.052) and MAE (0.033), the best MAPE (5.63%), and does so without overfitting. The strong performance of a well-regularised linear model is a healthy sign: the engineered features and PCA have produced a feature space where SOH is largely a linear function of the input measurements. The bias-variance tradeoff plot (training polynomial regression with degrees 1 through 5): training error monotonically decreases with degree while test error begins to rise after degree 2.



Phase 2d: Supervised Learning - Classification

While regression gives a precise numerical estimate of SOH, in many operational scenarios, a binary signal is more actionable: is this battery healthy or degraded? The classification task labels each cycle as Healthy (SOH ≥ threshold) or Degraded (SOH < threshold).

Class Balance Verification

Before any modelling, the class distribution was verified:

Health Class	Count	Proportion
Healthy	1,476	53.3%
Degraded	1,293	46.7%
Imbalance Ratio	—	1.14:1

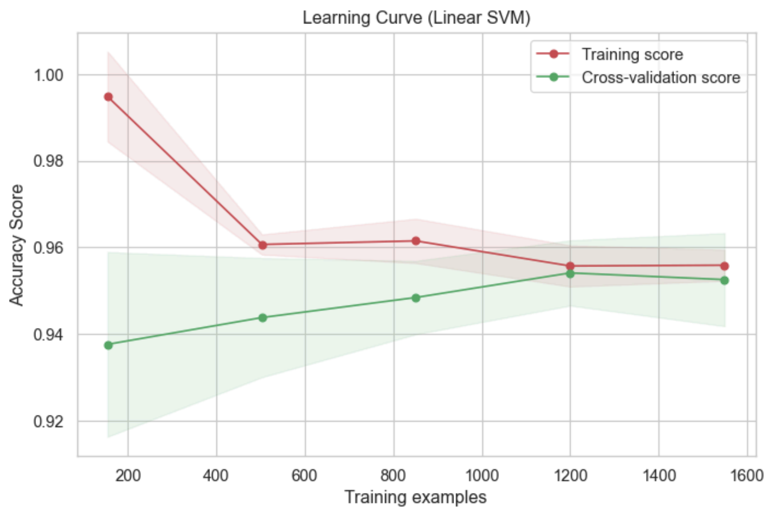
A 1.14:1 imbalance ratio is essentially balanced. No oversampling (SMOTE) or class weighting was needed. The near-equal split means accuracy is a meaningful metric (a naive majority classifier would only achieve 53.3%), and precision/recall can be interpreted without correction.

Model 1: Linear Classifier - Support Vector Machine (SVM)

SVM Model from scikit-learn was tuned over C = [0.01, 0.1, 1, 10, 100]. GridSearchCV with 5-fold cross-validation selected C=100, reflecting that tighter margins gave better generalisation on this relatively well-separated dataset.

Metric	Sklearn SVM (Test)
Accuracy	95.4%
Precision	97.2%
Recall	94.1%
F1-Score	95.7%
AUC-ROC	0.98

The performance gap between the custom and sklearn implementations is striking. The SVM, using the SMO algorithm, achieves high performance on both metrics (95.4% accuracy, 97.2% precision, 94.1% recall, 95.7% F1). The learning curve analysis showed that training and cross-validation accuracy converge at high values as the dataset grows, indicating an ideal bias-variance tradeoff with no evidence of severe overfitting.



Model 2: Non-Linear Classifier: Artificial Neural Network

- A Multi-Layer Perceptron (MLP) was trained using backpropagation. The design philosophy was deliberately conservative: a single small hidden layer to limit model complexity, L2 regularisation (alpha parameter), and early stopping to prevent memorisation. This approach trades a small amount of peak accuracy for much better generalisation to unseen batteries.

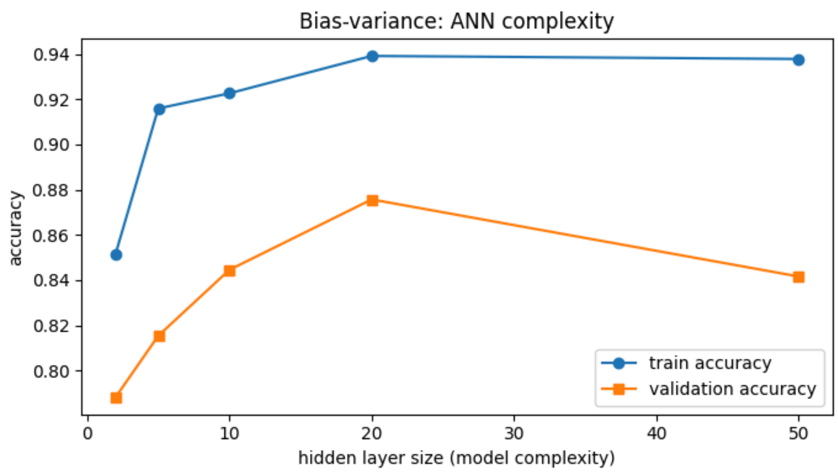
- Hyperparameter search via 5-fold GroupKFold cross-validation explored hidden layer sizes of (5,) and (10,) neurons with alpha values of [0.1, 1, 10, 100]. The selected configuration was:

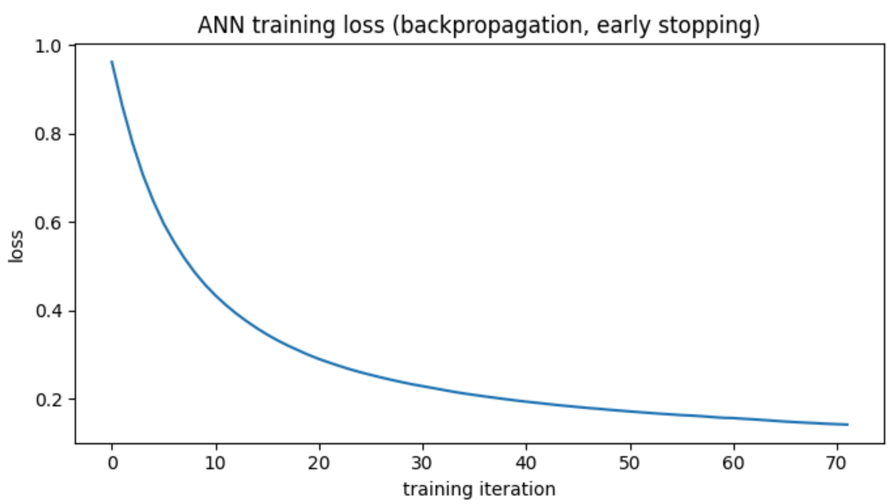
- Hidden layer: (5,) neurons
- Alpha (L2): 0.1
- Early stopping: enabled (15 iterations patience)
- Max iterations: 1,000

- Results:

Metric	ANN — Test Set
Accuracy	93.6%
Precision (Healthy)	98%
Recall (Healthy)	91%
F1-Score (Healthy)	95%
Precision (Degraded)	88%
Recall (Degraded)	97%
Misclassification Rate	6.44%
Train Accuracy	95.7%
CV Accuracy	91.6%

The ANN achieves 93.6% test accuracy, whereas the SVM achieved (95.4%). The ANN shows balanced precision and recall across both classes, a sign that the decision boundary is well-calibrated. The gap between train accuracy (95.7%) and CV accuracy (91.6%) is moderate, and the model was correctly assessed as 'balanced — no overfitting.' The bias-variance complexity plot (hidden layer size vs. accuracy) showed that accuracy peaks early at small network sizes and then plateaus or slightly degrades for larger networks, confirming that aggressive regularisation is keeping the network honest. The loss curve from early stopping shows clean convergence without oscillation.





Overall Classification Comparison

Model	Test Accuracy	F1-Score
Sklearn Linear SVM (C=100)	95.4%	95.7%
ANN/MLP (5 neurons, α=0.1)	93.6%	94% (avg)

Models Selection

The choice of models:

- **Ridge Regression** was chosen for SOH prediction because the statistical analysis showed that capacity (and hence SOH) has a smooth, monotonic relationship with many electrical features, suggesting linearity would suffice. It did.
- **Linear SVM** was chosen for classification because the PCA scatter plot showed two largely separable clusters, suggesting a hyperplane boundary would work well.
- **ANN** was added as a non-linear alternative to capture any boundary irregularities that a linear SVM might miss. The battery-specific degradation trajectories identified by HDBSCAN suggested the true decision boundary might have local curvature.
- **K-Means and HDBSCAN** were used to validate the structure of the dataset before committing to supervised labels, a step that bridges exploratory analysis and model training.

Summary of Results

Task	Best Model	Best Metric
SOH Regression	Ridge Regression (α=100)	Test R^2 = 0.88, MAPE = 5.63%
Health Classification	Sklearn SVM (C=100)	Test Accuracy = 95.4%, F1 = 95.7%
Unsupervised Validation	K-Means (K=2)	Silhouette = 0.25, confirms binary structure
Dimensionality Reduction	PCA (16 components)	95% variance retained from 30 features

Conclusion

This project showed that classical machine learning and neural network approaches can effectively predict lithium-ion battery health from cycle-level electrical measurements. By working through the problem systematically, from statistical validation, through unsupervised structure discovery, to supervised modelling, each phase informed the next. The ANOVA and t-tests confirmed that both temperature and health class have real, measurable effects on capacity, giving the modelling phase a solid statistical foundation. Clustering independently confirmed the Healthy/Degraded split is not an arbitrary label but a genuine pattern in the data. For regression, Ridge achieved an R^2 of 0.88 with a MAPE of just 5.63%, proving that a well-regularised linear model is all you need when the feature engineering is done right. On the classification side, SVM edged out the ANN with 95.4% accuracy and a 0.98 AUC-ROC, though both models performed strongly. The main takeaway is that thoughtful preprocessing, dropping uninformative columns, engineering physically meaningful features, and reducing dimensionality through PCA contributed more to final performance than model complexity ever could.

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